KAIZEN MATHS

Lesson starter

Advanced Factorisation: Grouping, Difference of Squares and Cubics

Complete Factorisation

Name _____

S1

$x^2 - 13x + 42$

S2

$10x^2 - 29x + 21$

S3

$15m^2 - 9m - 25m + 15$

S4

$5x^3 + 45x^2 + 100x$

Show your working

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Exit ticket

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E1

$x^3 - x^2 - 30x + 72$

E2

$x^2 - 9x + 20$

E3

$8x^2 - 26x + 21$

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Teacher answers & working

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S1

$x^2 - 13x + 42$

Answer:

$(x - 6)(x - 7)$

Working:

Method focus: find two numbers that multiply to c and add to b.

Step 1: Find two numbers that multiply to 42 and add to -13.

Step 2: The numbers are -6 and -7.

$x^2 - 13x + 42 = (x - 6)(x - 7)$

Step 3: Check by expanding the two brackets.

S2

$10x^2 - 29x + 21$

Answer:

$(2x - 3)(5x - 7)$

Working:

Method focus: split the middle term, then factorise by grouping.

Step 1: Multiply a × c for the quadratic: 10 × 21 = 210.

Step 2: Split the middle term using -14x and -15x, because they add to -29x.

$10x^2 - 29x + 21$

$= 10x^2 - 14x - 15x + 21$

Step 3: Group the four terms in pairs and factor each pair.

$= (10x^2 - 14x) - (15x - 21)$

$= 2x(5x - 7) - 3(5x - 7)$

Step 4: Factor out the shared bracket.

$= (2x - 3)(5x - 7)$

Step 5: Check by expanding.

Teacher answers & working

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S3

$15m^2 - 9m - 25m + 15$

Answer:

$(3m - 5)(5m - 3)$

Working:

$15m^2 - 9m - 25m + 15$

$= (15m^2 - 9m) - (25m - 15)$

$= 3m(5m - 3) - 5(5m - 3)$

$= (3m - 5)(5m - 3)$

S4

$5x^3 + 45x^2 + 100x$

Answer:

$5x(x + 5)(x + 4)$

Working:

Method focus: take out the common factor before factorising the quadratic inside.

Step 1: Take out the common factor 5x.

$5x^3 + 45x^2 + 100x = 5x(x^2 + 9x + 20)$

Step 2: Factorise the quadratic inside the bracket.

The numbers 5 and 4 multiply to 20 and add to 9.

$5x(x^2 + 9x + 20) = 5x(x + 5)(x + 4)$

Step 3: The expression is factorised completely.

E1

$x^3 - x^2 - 30x + 72$

Answer:

$(x - 4)(x - 3)(x + 6)$

Working:

Method focus: find one linear factor, then factorise the quadratic quotient.

Step 1: Test a simple root. Substituting x = 4 gives 0, so (x - 4) is a factor.

Step 2: Divide by (x - 4) to get the quadratic quotient.

$x^3 - x^2 - 30x + 72 = (x - 4)(x^2 + 3x - 18)$

Step 3: Factorise the quadratic quotient.

$(x - 4)(x^2 + 3x - 18) = (x - 4)(x - 3)(x + 6)$

Step 4: The cubic is now factorised completely.

Teacher answers & working

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E2

$x^2 - 9x + 20$

Answer:

$(x - 4)(x - 5)$

Working:

Method focus: find two numbers that multiply to c and add to b.

Step 1: Find two numbers that multiply to 20 and add to -9.

Step 2: The numbers are -4 and -5.

$x^2 - 9x + 20 = (x - 4)(x - 5)$

Step 3: Check by expanding the two brackets.

E3

$8x^2 - 26x + 21$

Answer:

$(2x - 3)(4x - 7)$

Working:

Method focus: split the middle term, then factorise by grouping.

Step 1: Multiply a × c for the quadratic: 8 × 21 = 168.

Step 2: Split the middle term using -14x and -12x, because they add to -26x.

$8x^2 - 26x + 21$

$= 8x^2 - 14x - 12x + 21$

Step 3: Group the four terms in pairs and factor each pair.

$= (8x^2 - 14x) - (12x - 21)$

$= 2x(4x - 7) - 3(4x - 7)$

Step 4: Factor out the shared bracket.

$= (2x - 3)(4x - 7)$

Step 5: Check by expanding.